Singular events, such as the births and deaths of individuals can make it difficult to directly perceive any probabilities that are hidden in a plethora of other circumstances. George R. Price[1] modelled selection as a truly mechanistic process. In particular his covariance formulation of the mathematical theory of Natural Selection is regarded as conceptually elucidating (Price, 1970).

We assume a single locus with two alleles, A and a, and calculate the change in frequency Q of the gene A from one generation to the next, ΔQ. In this case his equation looks like this:

$$\Delta Q = \frac{\mathrm{cov}(z,q)}{\overline{z}} \tag{1}$$

Here z_i is the number of successful gametes from individual i ($i = 1 \ldots N$ where N is the population size) and q_i is the frequency of A in **that** individual, $q_i = 0, 1/2, 1$. There is a significant part of individual i's life embedded in the parameter z_i. There is first and foremost the probability that he or she survives to breed, is fertile and finally the number of offspring.[2] $\overline{z}$ and $\overline{q}$ are the arithmetic means of the number of successful gametes and frequencies of A over all the N *individuals* in the population, respectively. $\mathrm{cov}(z,q)$ is the covariance between z_i and q_i

$$\mathrm{cov(z, q)} = \frac{\sum_{i=1}^{N}(z_i - \overline{z})(q_i - \overline{q})}{N} = \frac{\sum_{i=1}^{N} z_i q_i}{N} - \overline{z}\,\overline{q} \tag{2}$$

Note that Price adds an alternative formulation

$$\Delta Q = \frac{\beta_{zq}\sigma_q^2}{\overline{z}} \tag{1'}$$

Here σ_q^2 is the variance of q_i and β_{zq} is the slope of a regression of z_i, the successful gametes on q_i, the frequency of gene A in individual i. Here we can see how natural selection operates using standard statistical analysis. The sign of the change in gene frequency ΔQ is given by the slope β_{zq}. That is, when the composition of the population is such that z_i decreases with q_i ($\beta_{zq} < 0$) the gene frequency declines and the opposite is the case when $\beta_{zq} > 0$. I cannot think of a more clear example of 'the concept of a field of forces—introduces a dispositional physical entity described by certain equations (rather than by metaphors), in observable accelerations' (Popper, 1959, p. 30). I give a hypothetical example in Figure 5.

Price considers two populations P_1 and P_2, where the former represents the parents and the latter their offspring. We first consider the P_1. Let g_i be the dose of the gene A in individual i. $g = 0, 1, 2$ for a diploid species. Further, let n_z be the zygotic ploidy of the species. For a diploid species $n_z = 2$. The frequency of the gene A in P_1 is

$$Q_1 = \frac{\sum_{i=1}^{i=N} g_i}{n_z N} = \frac{\sum_{i=1}^{i=N} n_z q_i}{n_z N} = \overline{q} \tag{3}$$

where we take the sums over all members of population P_1.

In traditional population genetics you do this differently. Here you assume (i) Mendel's law of inheritance and (ii) random mating. (Price assumes nothing of this kind when deriving

[1] Price was American population geneticist and before that he had a career as a physical chemist and as science journalist. He worked in the Manhattan project with the physical and chemical properties of Uranium 235, contracted thyroid cancer during the 1950s. He died by his own hand 1975, possibly induced by hypothyroidism which is both a physical and psychiatric condition following thyroid hormone deficiency. He lived among homeless in London and gave away all his possessions, while working on the evolution of altruism at the Galton Laboratory, University College, London (Harman, 2011).

[2] Personally I have produced just two successful gametes, in spite of the fact that there are millions of sperms per ejaculation.

his equation.) Imagine that we pick up each individual and take a note of its genotype, if it is *AA*, *Aa* or *aa*. From those data we can easily calculate the gene frequency. Now we use the traditional assumptions and Eq. (3) to calculate $\overline{q}$. We knew the frequency Q_1 of A in the population P_1. Let us now form the population P_1 as zygotes from their parents' gametes. This is an urn experiment. We draw two A with probability Q_1^2, it follows from the multiplication theorem, Eq. (P3). Likewise, we pick up one A followed by a a with the probability $Q_1(1-Q_1)$ and one a followed by a A with probability $(1-Q_1)Q_1$—both using the multiplication theorem Eq. (P3) and then we sum them together per Eq. (P2). Since the two are identical, sum them together and get $2Q_1(1-Q_1)$.

$$Q_1 = \frac{2Q_1^2 N + 2Q_1(1-Q_1)N}{2N} = \overline{q} \tag{3'}$$

Here we have used 2 for g_i for all those that are homozygous and 1 for those that are heterozygous. This way to go from gene frequencies to genotype ones is called Hardy-Weinberg's law:

$$\begin{aligned} AA&: \quad Q^2 \\ Aa&: \quad 2Q(1-Q) \\ aa&: \quad (1-Q)^2 \end{aligned}$$

There is nothing more than Popper's theorems (or the corresponding traditional ones) behind this exercise. In **Appendix A** we will derive Price's equation Eq. (1) from first principles. Also, we use Popper's theorems to derive a dynamical system which makes it possible to calculate the change of gene frequencies over time. From Eq. (A3) in Appendix A.

$$\Delta Q = Q_2 - Q_1 = \frac{\mathrm{cov}(z,q)}{\overline{z}} + \frac{\sum_{i=1}^{N} z_i \Delta q_i}{N\overline{z}} \tag{4}$$

The second term in Eq. (4) is not a bug but a feature. It is the mean value of the differences between the frequency of A genes per successful gamete from individual i and the frequency of that gene in that individual. This is a gross simplification; we assume that all individuals with the same genotype are exactly the same in all other aspects. We assume that all those other aspects somehow even out. The genotypes have become what von Mises refers to as *classes* or *collectives* (von Mises, 1957).

Another factor that will make it non-zero is genetic drift, also known as sampling error. There are also a number of other factors, like sexual selection (Fisher, 1930, p. 146–156). Selection can also operate through meiotic drive and segregation distortion, when a gene is trying to get an upper hand in the Battle for Transmission (Lindholm *et al.*, 2016) rather than in improving survival and reproduction in general. That term is an argument for the propensity interpretation of probability.

The traditional population genetics uses another formulation. Remember the definition of cov and its variations, Eq. (2). From there we get

$$\mathrm{cov(z,q)} = \frac{\sum_{i=1}^{N} z_i q_i}{N} - \overline{z}\,\overline{q} = \frac{NQ_1^2 z_{AA} + N2Q_1(1-Q_1)z_{Aa}\,\frac{1}{2}}{N} - \overline{z}\,\overline{q}$$

where we assume Hardy-Weinberg equilibrium and remember the definition of q_i above. Also we replace the index i on the z_i (the number of successful gametes from individual i) with their genotypes. I also replace Q_1 and $\overline{q}$ with the letter p (they are equal under the given assumptions)

$$\mathrm{cov(z,q)} = p^2 z_{AA} + p(1-p)z_{Aa} - \overline{z}\,\overline{q}$$

Substituting this into Eq. (1) (using the p rather than Q) we get

$$\Delta p = \frac{p^2 z_{AA} + p(1-p) z_{Aa}}{\overline{z}} - p \quad (1'')$$

which is derived from the Price equation rather than the conventional way, as Okasha (2024) does it.

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